



**Department of Electronics and Communication Engineering
Khulna University of Engineering & Technology**

Proposal of Analog Electronics II Laboratory (ECE 2102)

1.

Group: B1	
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2. **Title:** Design and Implementation of a Heartbeat Detection System Using Operational Amplifier

3. **Background and Motivation: (150 – 200 words):**

The human heartbeat is one of the most important physiological signals used to monitor health conditions. A heartbeat detector is a useful biomedical electronic device that can measure pulse rate and help in the early detection of irregular heart activity. In recent years, low-cost and portable health monitoring systems have gained significant importance due to the increasing awareness of personal healthcare and preventive medicine.

This project aims to design and implement a heartbeat detector circuit using operational amplifiers (Op-Amps). The circuit will detect the tiny electrical or optical pulse signals generated by blood flow in the human body and amplify them into a measurable form. Different analog electronic components such as sensors, filters, amplifiers, and LEDs/buzzers will be used to process the heartbeat signal accurately.

The project will provide practical knowledge about analog signal conditioning, amplification, filtering, and biomedical instrumentation. It will also help students understand the real-life application of operational amplifiers in healthcare electronics. The developed system is expected to be simple, low-cost, power efficient, and suitable for educational and basic medical monitoring purposes.

4. Objectives (in listed form):

- To design a heartbeat detector circuit using Op-Amps.
- To amplify weak heartbeat signals using analog amplifiers.
- To filter unwanted noise from the detected signal.
- To display heartbeat indication using LED or buzzer output.
- To measure and observe pulse signals accurately.
- To gain practical experience in biomedical analog electronics.

5. Conflicting Requirements:

- High amplification may also amplify unwanted noise.
- Accurate signal detection requires proper filtering, which may increase circuit complexity.
- Low power consumption and high sensitivity are difficult to achieve simultaneously.
- Compact circuit design may reduce ease of troubleshooting and maintenance.

6. Proposed Budget for Implementing the Project:

Sl.	Item	Justification	Price (BDT)
1.	LM358 Operational Amplifier	Signal amplification and processing	45
2.	Infrared LED and Photodiode	Heartbeat signal detection	40
3.	Resistors (100Kohm, 10Kohm, 2.4Kohm, 330ohm, 100ohm) and 10K Potentiometers	Biasing, gain control, and filtering	50
4.	Red LEDs	Heartbeat indication	2
5.	9V Battery	Power supply for the circuit	75
6.	Breadboard and Connecting Wires	Circuit implementation and testing	280
7.	Finger Clip, Black Tape, and Miscellaneous Materials	Sensor attachment and assembly	150
Total (BDT)			642
In words: Six Hundred Forty-Two			

7. Work Plan (Week wise plan):

1 st Week	• Study Op-Amp and heartbeat detection principles • Design the heartbeat detector circuit
2 nd Week	• Build sensor and amplification stages • Test signal detection and amplification
3 rd Week	• Complete filtering and output stages • Test full system and prepare final report

8. Impact of project on the society, environment and sustainability:

The proposed heartbeat detector can contribute to low-cost healthcare monitoring systems, especially in rural and remote areas where medical facilities are limited. The project promotes awareness of personal health monitoring and encourages the development of affordable biomedical devices. Since the system uses low-power analog components, it is energy efficient and environmentally friendly. The project also supports sustainable engineering practices by utilizing inexpensive and reusable electronic components for educational and healthcare purposes.

Signature and Name of the Supervisor